
Considerations for Interpreting Packet Loss Observed by Active Performance Measurements

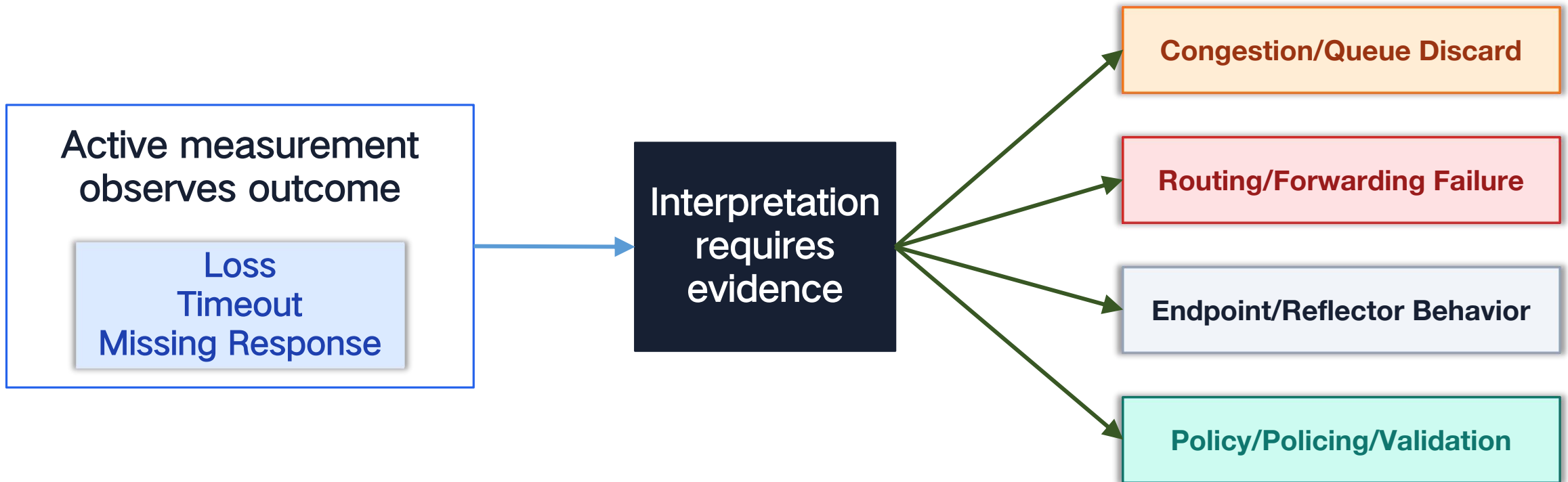
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Problem: the observation is clear, the cause is not

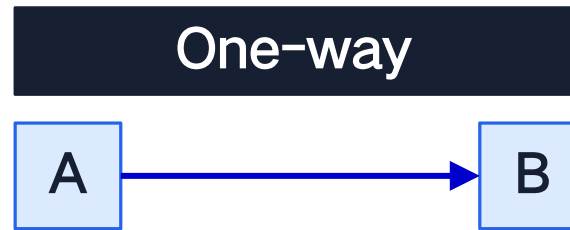
- ❑ A packet counted as lost is still lost from the measurement perspective, but it may mean different operational things.



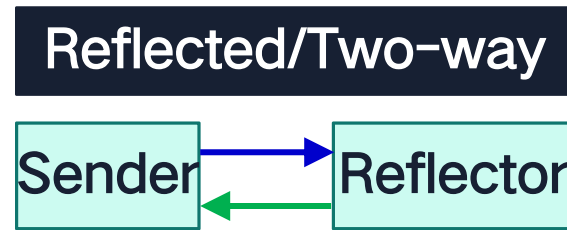
Same Observable Result $\neq$ Same Operational Response

Where Ambiguity Shows up in the Measurements

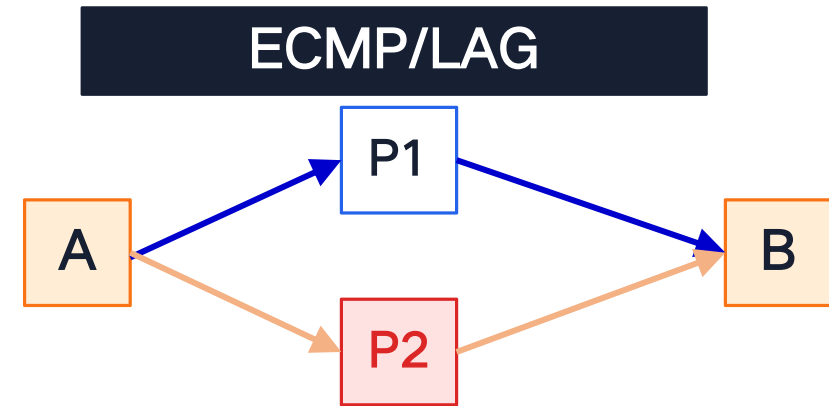
- ❑ One-way, reflected, and multipath measurements can all produce similar symptoms.



Direction is known;
discard point and cause
may be unknown.



Missing response may
involve forward path,
reflector behavior, or
return path.

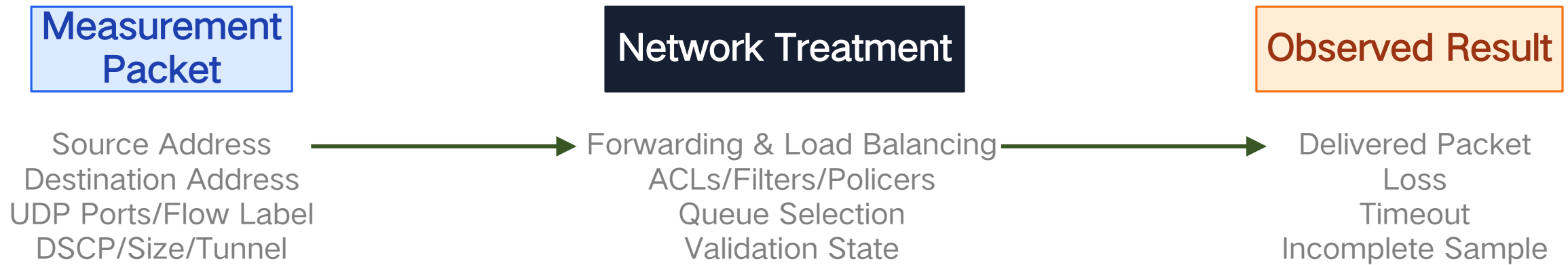


A different flow identifier can
exercise a different component path.

Implication: sender-side loss ratio, timeout, or delay samples should normally be reported with direction, endpoint role, packet/flow fields, and timing context.

Active Measurement Traffic May Receive Different Treatment

- ❑ Packet fields and validation context can change forwarding, filtering, policing, and path selection.



Validation-related discard is one plausible cause when evidence fits the pattern.

But validation-related function deployment alone is not evidence that validation caused the loss.

Proposed Interpretation Checklist

□ Keep the measured outcome distinct from the operational hypothesis.

1

Report the Observable Outcome

Lost Packet/Missing Response/Timeout/Delay Sample

2

Separate Inferred Cause

State the Evidence before Naming Congestion, Filtering, Validation, etc.

3

Check Direction and Role

One-way vs Reflected; Sender, Receiver, Reflector

4

Check Flow, Path, and Timing

Headers, ECMP/LAG Outcome, Transient Events

5

Check Representativeness

Does Test Traffic Match the Traffic Being Characterized?

Operational context that helps

Measurement Report

Timestamps, Sequence Numbers, Packet Fields

Routing/Forwarding State

Route Changes, Next Hops, Tunnels, ECMP

Counters and Logs

Interfaces, Queues, Policers, Filters, Rules

Endpoint/Reflector State

Session Active, Packet Received, Response Generated

OAM/Telemetry/Operator Context

Event Correlation and Administrative Boundaries

Document Scope and Requested Feedback

- ❑ The draft is an interpretation and reporting guidance document for network operators and measurement systems.

This Document DOES

- Provide common considerations for interpreting active-measurement loss
- Discuss one-way, reflected, delay, delay variation, and multipath implications
- Identify useful operational context and reporting practices

This document DOES NOT

- Define new metrics or active-measurement protocols
- Assign a single cause to every observed loss event
- Weaken filtering, policing, validation-related functions, or other security policies

Feedback wanted from WG

- Is the problem useful for network operators?
- Are the protocol-specific implications correct?
- What reporting/context guidance should be added or removed?

Next Step

- Refine the draft based on the feedback

Thanks!